1 Deriving Continuity Equation

Beginning with a control volume, as shown below, it is possible to derive a continuity equation for mass balance.

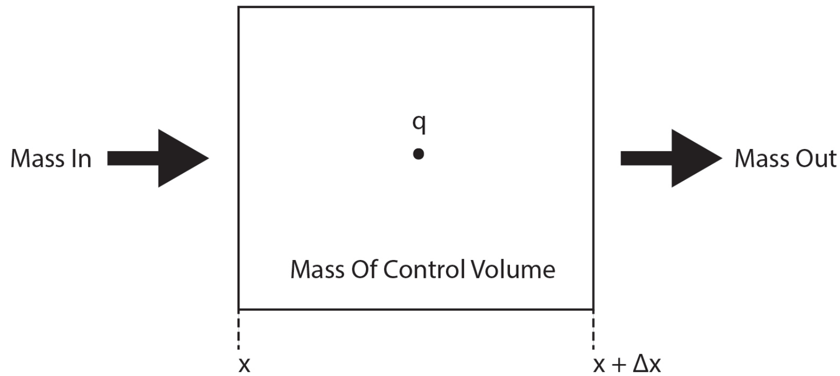

Figure 1: Control volume defined to develop mass balance equation.

Because mass can neither be created or destroyed, the mass conservation for this control volume can be described as

$$\underbrace{\text{Mass Final} - \text{Mass Initial}}_{\text{Accumulation}} = \underbrace{\text{Mass In} - \text{Mass Out}}_{\text{Convection}} + \underbrace{\text{Generation}}_{\text{Source}}, \tag{1}$$

which can be stated as

$$\underbrace{m \cdot V\Big|_{t+\Delta t} - m \cdot V\Big|_{t}}_{\text{Accumulation}} = \underbrace{F \cdot \Delta t \cdot A\Big|_{x} - F \cdot \Delta t \cdot A\Big|_{x+\Delta x}}_{\text{Convection}} + \underbrace{q \cdot \Delta t \cdot V}_{\text{Source}}. \tag{2}$$

Here, m is the mass density of the control volume, F is the mass flux, V is the volume of the control volume, Δt is the change in time, and A is the cross-sectional area orthogonal to the direction of Δx. q is in units of $\frac{\text{kg}}{\text{m}^3 \cdot \text{s}}$. By dividing this equation by the Δt and V, one obtains

$$\frac{m}{\Delta t}\Big|_{t+\Delta t} - \frac{m}{\Delta t}\Big|_{t} = \frac{F}{\Delta x}\Big|_{x} - \frac{F}{\Delta x}\Big|_{x+\Delta x} + q, \tag{3}$$

which can be reformulated as

$$\frac{m\Big|_{t+\Delta t} - m\Big|_{t}}{\Delta t} = \frac{F\Big|_{x} - F\Big|_{x+\Delta x}}{\Delta x} + q. \tag{4}$$

Next, the limit is taken as Δt and Δx go to zero, i.e.,

$$\lim_{\substack{\Delta t \to 0 \\ \Delta x \to 0}} \left(\frac{m\Big|_{t+\Delta t} - m\Big|_{t}}{\Delta t} = \frac{F\Big|_{x} - F\Big|_{x+\Delta x}}{\Delta x} + q \right), \tag{5}$$

resulting in

$$\frac{\partial m}{\partial t} = -\frac{\partial F}{\partial x} + q. \tag{6}$$

Note that this equation, representing the conservation of total mass, is also called the Continuity Equation. In general form, the equation can be stated as

$$\frac{\partial m}{\partial t} = -\nabla \cdot F + q. \tag{7}$$

The m (mass density of fluid per bulk volume) of the left term can be replaced like

$$\frac{\partial (\phi\rho)}{\partial t} = -\nabla \cdot F + q, \tag{8}$$

where ϕ is porosity and ρ is fluid density. The source term q represents mass injection or production by wells. Positive values correspond to injection wells, while negative values correspond to production wells. The F (mass flux in units of $\frac{kg}{m^3}\frac{m}{sec}$) can be replaced like

$$\frac{\partial (\phi\rho)}{\partial t} = -\nabla \cdot (\rho \cdot u) + q, \tag{9}$$

where u is fluid velocity. Next, by assuming that fluid transport is entirely governed by pressure-driven advection (diffusion is neglected), and that fluid velocity can be described by Darcy's Law, $u = -\frac{k}{\mu}\nabla P$, where k is permeability and μ fluid viscosity,

$$\frac{\partial (\phi\rho)}{\partial t} = -\nabla \cdot \left(\rho\left(-\frac{k}{\mu}\nabla P\right)\right) + q. \tag{10}$$

Considering that viscosity is constant and that porosity is constant in time, the continuity equation is,

$$\phi\frac{\partial \rho}{\partial t} - \frac{1}{\mu}\nabla \cdot (\rho k \nabla P) = q. \tag{11}$$

2 Finite Volume Method Discretization

Beginning with the continuity equation from mass balance, Equation 11,

$$\phi\frac{\partial \rho}{\partial t} - \frac{1}{\mu}\nabla \cdot (\rho k \nabla P) = q, \tag{12}$$

the equation is integrated over a control volume V_i,

$$\int_{V_i}\left[\phi\frac{\partial \rho}{\partial t} - \frac{1}{\mu}\nabla \cdot (\rho k \nabla P)\right] dV = \int_{V_i} q dV. \tag{13}$$

Applying the divergence theorem converts the volume integral into a surface integral,

$$\phi V_i\frac{\partial \rho}{\partial t} - \frac{1}{\mu}\int_{A_i} \rho k \nabla P \cdot \mathbf{n}\, dA = Q, \tag{14}$$

where $\mathbf{n}$ is the outward-pointing unit normal vector. Note that $Q = qV_i$ is the mass injection rate in units of $\frac{\text{kg}}{\text{s}}$. The volume of the cell, $V_i = A\Delta x$, means this can be discretized as,

$$\phi_i A \Delta x \frac{\rho_i^\nu - \rho_i^n}{\Delta t} - \frac{A}{\mu}\left[\rho_{i+\frac{1}{2}}^\nu k_{i+\frac{1}{2}} \frac{P_{i+1}^\nu - P_i^\nu}{\Delta x} - \rho_{i-\frac{1}{2}}^\nu k_{i-\frac{1}{2}} \frac{P_i^\nu - P_{i-1}^\nu}{\Delta x}\right] - Q_i = r_i^\nu, \tag{15}$$

where n represents the previous time step, ν the current iteration, i the cell, Δt is the time step, and Δx is the cell length. Fluid densities in the convection term correspond to cell interfaces, e.g., $i+\frac{1}{2}$, and are upstreamed. Similarly, permeabilities also correspond to cell interfaces, with the harmonic average used, i.e., $k_{i+\frac{1}{2}} = \frac{2k_i k_{i+1}}{k_i + k_{i+1}}$, which guarantees continuity of flux across material interfaces. The residual, r, is defined for each cell. This equation is written for each cell and the system of all of the equations is solved fully implicitly as,

$$r^{n+1} = r^{\nu+1} \approx r^\nu + \frac{\partial r^\nu}{\partial P} \Delta P^{\nu+1} = 0, \tag{16}$$

such that the update to pressure at each iteration, $\Delta P^{\nu+1}$, is found as,

$$\frac{\partial r^\nu}{\partial P} \Delta P^{\nu+1} = -r^\nu. \tag{17}$$

The term $\frac{\partial r^\nu}{\partial P}$ represents the Jacobian. Note that it is defined for each iteration (as opposed to each time step) as the discretization is fully implicit. Once the pressure update is found and applied to create new cell pressures, the residual is rebuilt to check for convergence. Once convergence is achieved the simulator passes to the next time step.

Note that, under no-flow (Neumann) boundary conditions, the normal pressure gradient at the domain boundaries is zero, resulting in zero flux through the boundary faces. Therefore, the boundary flux terms do not contribute to the residuals of the edge cells (r_1 and r_{N_x}).

To illustrate how one might construct this system of equations, the following general example is given,

$$\begin{bmatrix} \frac{\partial r_1^\nu}{\partial P_1} & \frac{\partial r_1^\nu}{\partial P_2} & 0 & \cdots & 0 \\ \frac{\partial r_2^\nu}{\partial P_1} & \frac{\partial r_2^\nu}{\partial P_2} & \frac{\partial r_2^\nu}{\partial P_3} & \cdots & 0 \\ 0 & \frac{\partial r_3^\nu}{\partial P_2} & \frac{\partial r_3^\nu}{\partial P_3} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \frac{\partial r_{N_x}^\nu}{\partial P_{N_x-1}} & \frac{\partial r_{N_x}^\nu}{\partial P_{N_x}} \end{bmatrix} \begin{bmatrix} \Delta P_1 \\ \Delta P_2 \\ \Delta P_3 \\ \vdots \\ \Delta P_{N_x} \end{bmatrix} = - \begin{bmatrix} r_1^\nu \\ r_2^\nu \\ r_3^\nu \\ \vdots \\ r_{N_x}^\nu \end{bmatrix}. \tag{18}$$

2.1 Wells

Rate-prescribed wells

In the case that the rate of the well is to be prescribed, the term Q_i is simply added to the residual of the cell in which the well is located as show above. This well, or source term, has no explicit contribution to the Jacobian as it is a prescribed mass rate, independent of the pressure in the cell.

Pressure-prescribed wells

In the case that the pressure of the well is to be prescribed, the term Q_i is found as,

$$Q_i = WI \frac{\rho}{\mu} \left(P_{\text{inj}} - P_i\right), \tag{19}$$

where P_{inj} is the prescribed injection pressure and ρ is calculated based on P_{inj} (meaning its derivative with respect to the cell pressure is zero. WI is the well index and can either be prescribed or found as,

$$WI = \frac{2\pi kh}{\ln\left(\frac{r_e}{r_w}\right) + s}, \tag{20}$$

where h is a well height, r_e is an effective wellbore radius, r_w is the actual wellbore radius, and s is a skin factor. In the case of an injecting well, the derivative of the well term becomes,

$$\frac{\partial Q_i}{\partial P_i} = -WI\frac{\rho}{\mu}. \tag{21}$$

However, for a producing well, this derivative is,

$$\frac{\partial Q_i}{\partial P_i} = -WI\frac{\rho_i}{\mu} + \frac{WI}{\mu}\frac{\partial \rho_i}{\partial P_i}\left(P_{\text{inj}} - P_i\right). \tag{22}$$

In this sense, the fluid density is upstreamed.

2.2 Fluid density

For a slightly compressible fluid, the fluid compressibility is defined as

$$c_f = \frac{1}{\rho}\frac{\partial \rho}{\partial P}. \tag{23}$$

Assuming that c_f is constant (so that the fluid is slightly compressible),

$$\frac{1}{\rho}d\rho = c_f dP. \tag{24}$$

Integrating between a reference state (P_0, ρ_0) and the current state (P, ρ) gives

$$\int_{\rho_0}^{\rho} \frac{1}{\rho}d\rho = c_f \int_{P_0}^{P} dP, \tag{25}$$

which yields

$$\ln\left(\frac{\rho}{\rho_0}\right) = c_f(P - P_0). \tag{26}$$

Exponentiating both sides results in

$$\rho = \rho_0 e^{c_f(P-P_0)}, \tag{27}$$

which is used to calculate the fluid densities in each cell.

2.3 Fluid velocity

The fluid velocity across the interfaces can then be found using Darcy's Law as,

$$u = -\frac{k}{\mu}\frac{\partial P}{\partial x}. \tag{28}$$

Applying a finite-difference approximation at interface $i + \frac{1}{2}$ gives

$$u_{i+\frac{1}{2}} = -\frac{k_{i+\frac{1}{2}}}{\mu}\left(\frac{P_{i+1} - P_i}{\Delta x}\right), \tag{29}$$

with the mass flux $F_{i+\frac{1}{2}} = \rho_{i+\frac{1}{2}} u_{i+\frac{1}{2}}$. Note that for the domain boundaries the velocities are,

$$u_L = 0, \quad u_R = 0. \tag{30}$$

This is because the boundary condition used here is no-flow (Neumann).